



ST BENEDICT'S

SUBJECT	PHYSICAL SCIENCE: CHEMISTRY
GRADE	11
EXAMINER	Mr McCoy
MODERATOR	Mr Hilder
DATE	25 July 2023
MARKS	165
DURATION	2,5 hours

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LEARNER NAME:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of **20** pages and a separate data sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae, below, whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. Additional space is included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	Total
Mark									
Moderated Mark									
Question Total	16	21	24	31	8	21	27	17	165

Comment:

QUESTION 1**16 MARKS**

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	C	D
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Here the option C has been marked as an example.

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D

1.1 The chemical formula for magnesium sulphite is:

- A $\text{Mg}(\text{SO}_3)_2$
- B MgSO_4
- C Mg_3SO_2
- D MgSO_3

1.2 The chemical formula of ammonium dichromate is:

- A $(\text{NH}_4)_3\text{Cr}_2\text{O}_7$
- B $\text{NH}_4\text{Cr}_2\text{O}_7$
- C $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$
- D $\text{NH}_4(\text{Cr}_2\text{O}_7)_2$

1.3 The bonds between the atoms in the nitrate ion, NO_3^- can best be classified as:

- A Ionic bonds
- B Pure covalent bonds
- C Polar covalent bonds
- D London bonds

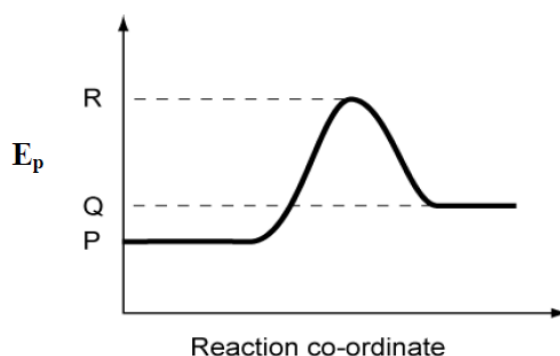
- 1.4 Which one of the following best describes the particles and how the electrons are involved in the process of ionic bonding?

	Particles	Electrons
A	Non-metal ions only	Transferred
B	Metal ions and non-metal ions	Shared
C	Non-metal ions only	Shared
D	Metal ions and non-metal ions	Transferred

- 1.5 Which of the following gives the correct number of atoms in 10 g of NH_3 (g)?

- A $\frac{10}{10} \times 6,02 \times 10^{23} \times 4$
 B $\frac{10}{17} \times 4$
 C $\frac{10}{17} \times 6,02 \times 10^{23} \times 4$
 D $\frac{10}{17} \times 6,02 \times 10^{23}$

- 1.6 The graph below shows the change in potential energy (E_p) as a function of the course of a reaction:

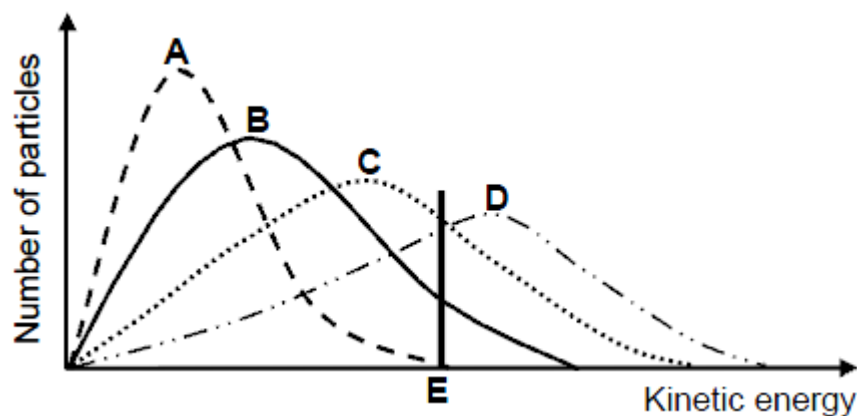


Which one of the following statements is correct?

- A The forward reaction is exothermic
 B The heat of the reaction for the forward reaction is given by $R - Q$
 C The activation energy for the forward reaction is given by $R - Q$
 D The heat of the reaction for the forward reaction is given by $Q - P$

- 1.7 The Maxwell-Boltzmann energy distribution curves below show the number of particles as a function of their kinetic energy for a reaction at four different temperatures. The minimum kinetic energy needed for effective collisions to take place is represented by E.

Which ONE of these curves represents the reaction with the highest rate?



- 1.8 A 5 kg bag of rice contains about $0,2 \times 10^6$ grains of rice. This means that average mass of each rice grain is:
- A $2,5 \times 10^{-5}$ g
 - B 25 000 μ g
 - C 0,025 kg
 - D 250 mg

QUESTION 2 CHEMICAL BONDING

21 MARKS

Iron and oxygen react to form iron (III) oxide, which has a high melting point.

- 2.1 Write a balanced equation to represent the reaction that takes place. Include state symbols. (4)

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- 2.2 Explain, in terms of its bonding and structure, why iron (III) oxide...

- 2.2.1 has a high melting point. (4)

2.2.2 does not conduct electricity when solid. (1)

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2.3 Iron conducts electricity both when solid and when molten. Describe the bonding structure of solid iron and explain why it conducts electricity. (4)

2.4 Iron (III) oxide has a melting point of 1565°C while iron (II) oxide has a melting point of 1377°C .

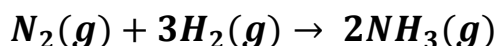
2.4.1 Use Lewis diagrams to show the formation of iron (II) oxide. (3)

2.4.2 Describe the bonding in iron (II) oxide. (2)

2.4.3 Explain why the two oxides would have different melting points. (3)

QUESTION 3 CHEMICAL BONDING**24 MARKS**

Nitrogen reacts with hydrogen to form ammonia as follows:



The intramolecular bonds found in nitrogen are **non-polar covalent bonds**.

3.1 Circle the correct word between brackets in the statement below:

This type of bond is found between the nitrogen (ATOMS / MOLECULE). (1)

3.2 Explain why the bond is called a **covalent** bond. (2)

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3.3 Explain why the bond is **non-polar**. (2)

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3.4 Ammonia is a polar molecule.

3.4.1 Determine the bond polarity. (3)

3.4.2 Hence, use a Couper diagram and an explanation to show why the molecule is polar. (4)

3.4.3 Ammonia liquefies at $-33,1\text{ }^{\circ}\text{C}$.

a) Name the specific type of intermolecular force in liquid ammonia. (1)

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b) Draw three molecules to show the arrangement of the intermolecular force above. (3)

3.5 Nitrogen gas liquefies at $-195,8\text{ }^{\circ}\text{C}$.

3.5.1 Name the specific type of intermolecular force in liquid nitrogen. (1)

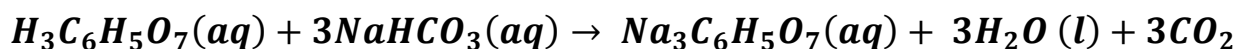
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3.5.2 Explain, in terms of the origin of the intermolecular forces, how nitrogen gas liquefies. (4)

3.6 Fully account for the difference in the liquefying temperatures of ammonia gas and nitrogen gas. (3)

QUESTION 4 ENERGY AND RATES**31 MARKS**

The reaction of citric acid and sodium bicarbonate produces sodium citrate and water in addition to the carbon dioxide, according to the following balanced chemical equation:



Consider the heat of reaction.

4.1 Define *heat of reaction*. (2)

4.2 Define *endothermic reaction*. (2)

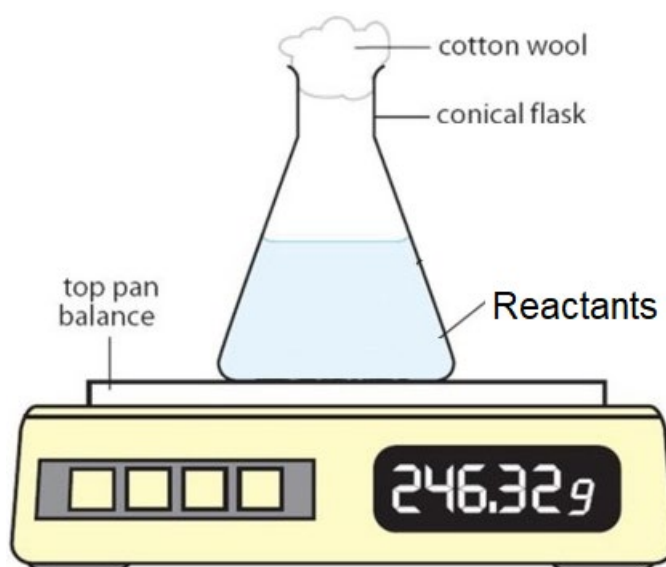
4.3 In the reaction above, more energy is absorbed than is released. Classify the reaction as ENDOTHERMIC or EXOTHERMIC. (1)

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4.4 In order for the reaction above to proceed, some bonds must be broken and others must be formed. Is bond breaking an ENDOTHERMIC or EXOTHERMIC process? (1)

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4.5 The diagram below shows the equipment that could be used to measure the rate of the reaction.



4.5.1 Which quantities are being measured and why is it possible? (3)

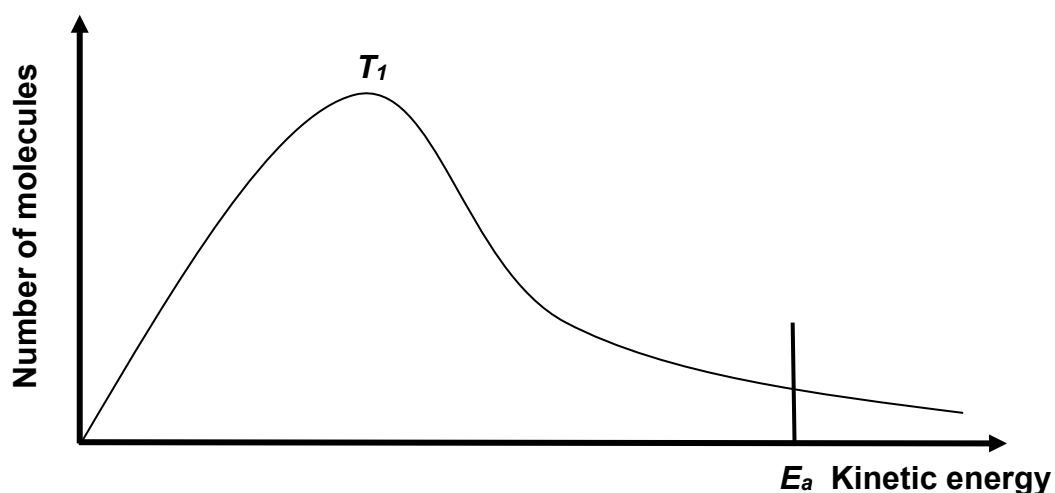
4.5.2 Why is cotton wool placed at the mouth of the conical flask? (3)

4.5.3 An additional piece of equipment is need to measure the rate of the reaction. Name the piece of equipment. (1)

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4.5.4 Name an alternative piece of equipment that could be used and the measured quantity. (2)

4.6 The graph below shows the Maxwell-Boltzmann distribution curve for the reaction and a given temperature, T_1 . The activation energy for the reaction, E_a , is marked.



4.6.1 On the graph above:

- Draw a new distribution curve for the same reaction at a lower temperature. Label this curve as T_2 . (2)
- Draw a new distribution curve for T_1 , if the concentration of reactants was increased. Label this curve as T_3 . (2)
- Indicate the position of the activation energy of the reaction at the lower temperature T_2 . Label this position as H . (2)

4.6.2 Define *activation energy*. (2)

4.6.3 Define *catalyst*. (2)

4.6.4 On the graph, indicated the position of the activation energy of the reaction when a catalyst is used. Label this position as **C**. (2)

4.6.5 Explain how the use of a catalyst results in reactions occurring at a faster rate. Refer to the Maxwell-Boltzmann curve and the collision theory. (4)

QUESTION 5**8 MARKS**

Consider the following two solutions in separate beakers:

Beaker A: Contains 250 cm³ of a 0,75 mol.dm⁻³ ammonium phosphate ((NH₄)₃PO₄) solution.

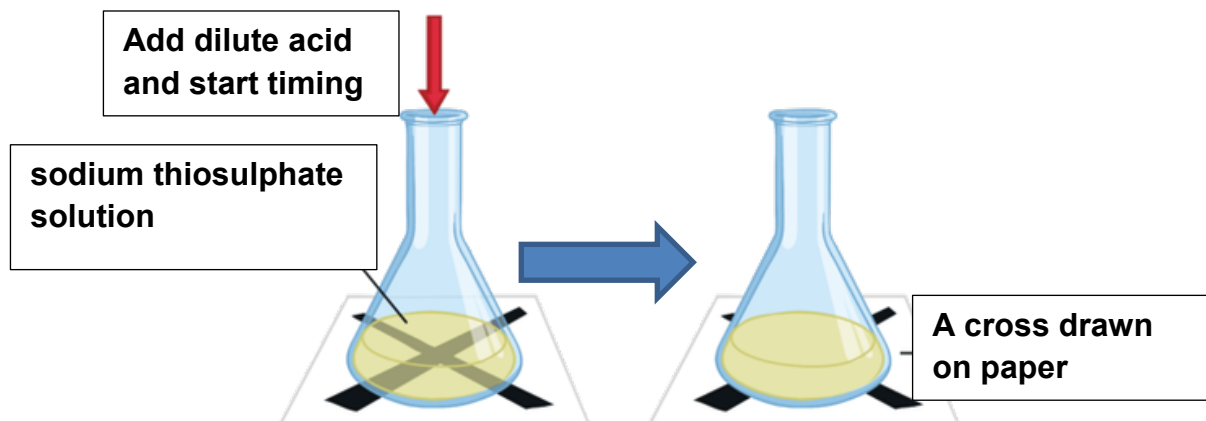
Beaker B: Contains 500 ml of a solution containing 20 g of ammonium nitrate (NH₄NO₃).

- 5.1 Write the dissociation reaction for ammonium phosphate, (NH₄)₃PO₄ and ammonium nitrate, NH₄NO₃. (4)

- 5.2 What will the concentration of the ammonium ions be if the contents of the two beakers are combined? (4)

QUESTION 6 ENERGY AND RATES**21 MARKS**

Priyesh wants to investigate the reaction between sodium thiosulphate solution and dilute hydrochloric acid.



Priyesh recorded the time taken for the reaction mixture to become too cloudy to see the cross. He used different concentrations of sodium thiosulphate solution combined with a fixed volume of dilute hydrochloric acid. Priyesh recorded data for six different concentrations of thiosulphate solution.

NOTE: All reactions are carried out in the same atmospheric conditions.

Priyesh's results are shown in the table below:

Concentration of $\text{Na}_2\text{S}_2\text{O}_3$ solution ($\text{mol}\cdot\text{dm}^3$)	Time taken for mixture to become too cloudy (s)				Reaction Rate ($\times 10^{-3} \text{ s}^{-1}$)
	Trial 1	Trial 2	Trial 3	Average	
0,10	126,2	124,7	125,4	125,4	7,97
0,15	81,4	83,8	84,5	83,2	12,02
0,25	49,2	49,3	49,6	49,4	20,24
0,40	31,3	31,4	31,2	31,3	31,95
0,45	27,2	27,9	28,4	27,8	35,97
0,50	23,5	24,8	26,7		

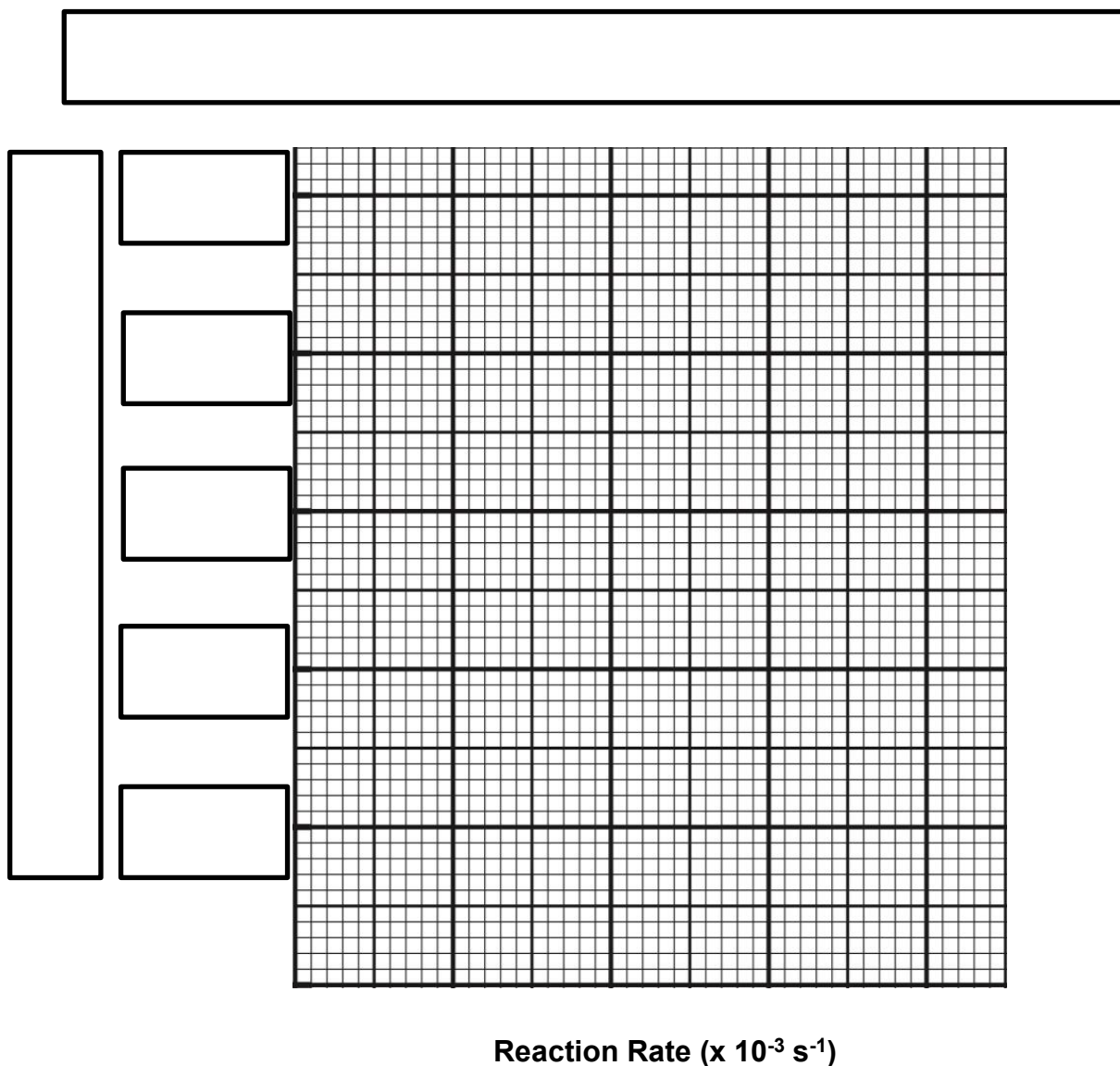
6.1 Refer to the table of results to answer these questions.

6.1.1 Calculate and write down the missing average time for the solution with a concentration of $0,50 \text{ mol}\cdot\text{dm}^3$. (2)

6.1.2 Write down the equation Priyesh used to calculate the Reaction Rate values. (2)

6.1.3 Complete the graph of concentration vs reaction rate below by:

- a) filling in the boxes to indicate the scale on the y-axis. (2)
- b) plotting the first five relevant points from the table and drawing the line of best fit. (5)

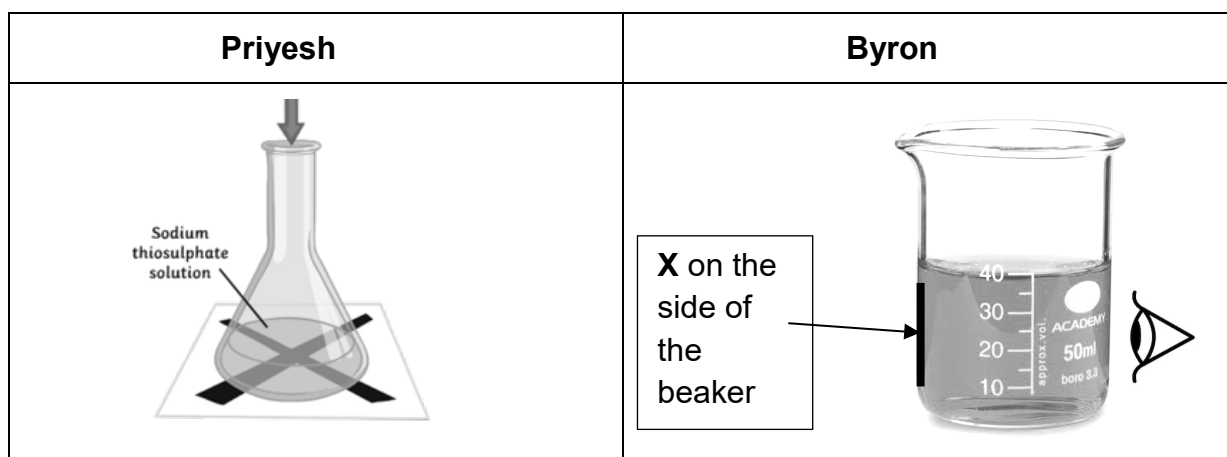


6.2 Priyesh carries out exactly the same reaction on a colder day. The reaction rate values calculated are different.

6.2.1 Draw a line on the axis above for the results he would obtain on a colder day. Label this line "Cold". (2)

6.2.2 Explain, in terms of the collision theory, the effect of a decrease in temperature on reaction rate. (4)

6.3 Byron and Priyesh use the same reactions on a third day but they each used slightly different apparatus and methods as shown in the diagrams below.



Both students timed how long it would take for the mark to "disappear" at 20 °C for two different experiments.

Experiment 2 was the same as Experiment 1 except that the volumes of all solutions were doubled.

	Experiment 1 time (s)	Experiment 2 time (s)
Priyesh	18	7
Byron	10	

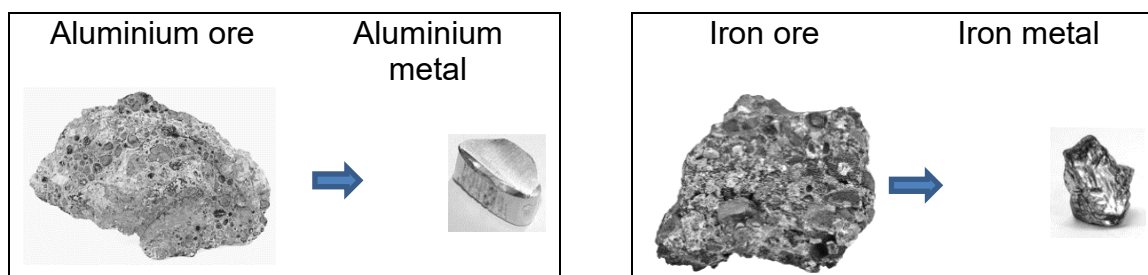
Given that they made NO errors in their experiments:

6.3.1 explain why **Priyesh's** time for Experiment 2 was lower than it was for Experiment 1. (2)

6.3.2 write down the value for **Byron's** Experiment 2. (2)

QUESTION 7 QUANTITATIVE CHEMISTRY**27 MARK**

A student was asked to compare the industrial processes used to extract aluminium and iron from their ores.



Aluminium is extracted from purified aluminium oxide.

7.1 Give the chemical formula for aluminium oxide. (2)

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The aluminium oxide reacts with hot concentrated sodium hydroxide and water to form a solution of sodium tetrahydroxoaluminate ($\text{NaAl}(\text{OH})_4$) according to the following balanced chemical equation:



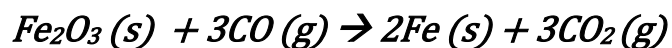
7.2 200 ml of a $2,0 \text{ mol}\cdot\text{dm}^{-3}$ solution of sodium hydroxide is reacted with excess aluminium oxide and 12 g of water:

7.2.1 Determine which reagent is limiting between the sodium hydroxide and the water. (4)

7.2.2 Explain the term *yield* in the context of a chemical reaction. (2)

- 7.2.3 If 45 g of $\text{NaAl}(\text{OH})_4$ is produced in this reaction, determine the percentage yield of $\text{NaAl}(\text{OH})_4$. (5)

In the extraction of iron from its ore, carbon monoxide is reacted with iron oxide according to the following balanced chemical equation:



- 7.3 A mass of 1 kg of Fe must be produced. Assume the reaction is 100% efficient and the carbon monoxide is in excess.

- 7.3.1 Determine the amount of iron oxide that is required. (3)

- 7.3.2 Determine the number of iron atoms that is required. (4)

- 7.3.3 Determine the volume of carbon dioxide that is produced if the reaction is carried out at STP. (3)

- 7.3.4 What mass of iron oxide would be required if the reaction was only 70% efficient? 1 kg of Fe must still be obtained. (4)

QUESTION 8 QUANTITATIVE CHEMISTRY**17 MARKS**

Niccolo preformed an experiment to determine the percentage of water of crystallization in hydrated sodium carbonate.

- 8.1 Explain what is meant by the term 'water of crystallization'. (2)

The water of crystallization is normally expressed in the chemical formula of the substance. The formula for hydrated sodium carbonate is $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$.

- 8.2 Explain what is meant by the 'x' in the formula. (2)

Niccolo used a procedure known as a titration to determine the percentage of water of crystallization in sodium carbonate. The hydrated sodium carbonate is reacted with a solution of hydrochloric acid (HCl) of known concentration.

The first step in this procedure is to prepare a $2 \text{ mol} \cdot \text{dm}^{-3}$ solution of HCl by diluting a concentrated HCl solution ($10 \text{ mol} \cdot \text{dm}^{-3}$).

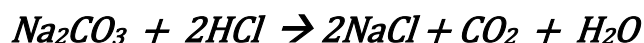
- 8.3 What is a solution of known concentration referred to as? (1)

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- 8.4 Calculate the volume of water that must be added to 200 ml of the $10 \text{ mol}\cdot\text{dm}^{-3}$ HCl solution to prepare a solution of HCl with a concentration of $2 \text{ mol}\cdot\text{dm}^{-3}$. (3)

Niccolo reacted 23,5 ml of the $2 \text{ mol}\cdot\text{dm}^{-3}$ hydrochloric with the hydrated sodium carbonate. All the dehydrated sodium carbonate reacts.

The balanced equation is shown below:



- 8.5 Determine the moles of dehydrated sodium carbonate that reacted. (3)

Niccolo correctly calculated that the mass of the dehydrated sodium carbonate that reacted is 2,491 g.

- 8.6 If the mass of the hydrated sodium carbonate was 5 g, determine the number of moles of water that is present in the water of crystallization. (3)

- 8.7 Hence, determine the value of x in the formula $\text{Na}_2\text{CO}_3\cdot x\text{H}_2\text{O}$. (3)

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE
ADDITIONAL SPACE TO ENSURE THAT ALL ANSWERS ARE MARKED

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This image shows a full page of graph paper. On the far left, there is a wide vertical margin. Within this margin, there are five identical, empty rectangular boxes stacked vertically, separated by small gaps. The rest of the page is filled with a standard grid pattern of thin black lines forming small squares. There are no markings or text on the grid itself.[illegible]

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